Lecture 7: Circuit Graphs, the Incidence Matrix and Tellegen's Theorem

EE201 – Circuit Theory I

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current voltage power nodes

Interconnection equations

A circuit with b branches: $2b$ unknowns, v_k and i_k for each branch k.

Group	How many	Describes
element equations	b	*what* each element is: $v = Ri$, a source value, ...
interconnection equations	b	*how* they are wired: $n-1$ KCL, $b-n+1$ KVL

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KCL and KVL do not care what the elements are

Replace a resistor by a capacitor or a source, keep the wiring: KCL and KVL stay the same. Keep only the wiring — the graph — and write all KCL equations as one matrix equation.

The graph of a circuit

Circuit graph

Replace every two-terminal element by a line segment, a branch. The points where branches meet are the nodes. The graph records only *which nodes each branch joins*; the shape of the drawing carries no information.

Procedure

1. Label each branch: $1, \ldots, b$.
2. Give each branch a direction — arbitrary.
3. Label each node: $\textcircled{1}$, $\ldots$, $\textcircled{n}$.
4. Draw the graph: dots and arrows.

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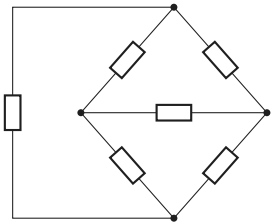
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The arrow is the reference direction of i_k. With associated references, v_k has $+$ at the tail.

If the current flows the other way, i_k is simply negative.

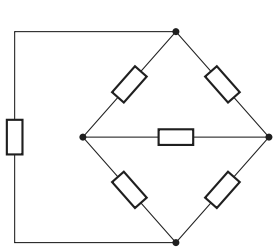
Example: the graph of a bridge circuit



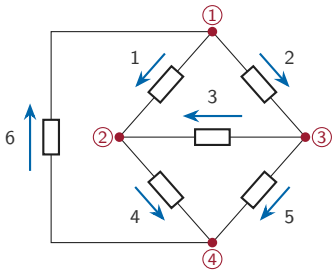
circuit $\mathcal{N}$

Six elements, drawn as boxes: their type does not matter here.

Example: the graph of a bridge circuit



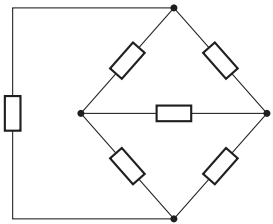
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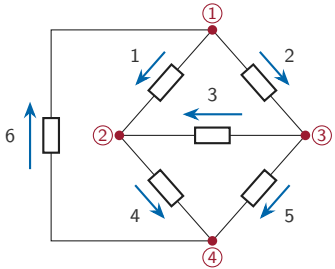
steps 1–3

Number the branches, give each an arrow (current reference), number the nodes.

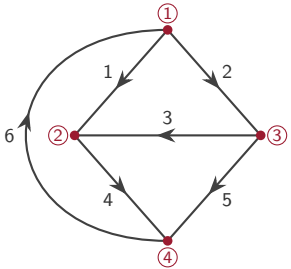
Example: the graph of a bridge circuit



circuit $\mathcal{N}$



steps 1–3



step 4: the graph

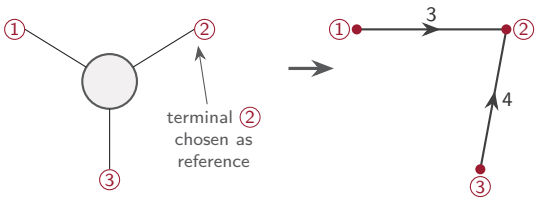
Nodes $\{①, ②, ③, ④\}$, $n = 4$; branches $\{1, \ldots, 6\}$, $b = 6$.

Branch 6 is curved only for clarity — a straight arrow is the same graph.

Elements with more than two terminals

A k-terminal element

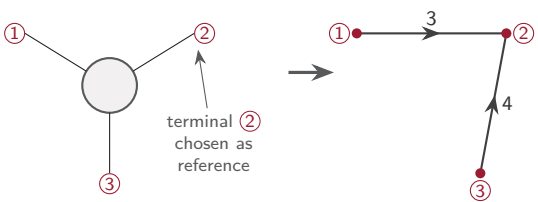
Choose one terminal as its reference (arbitrary). Draw $k-1$ branches, one from each other terminal to the reference terminal.



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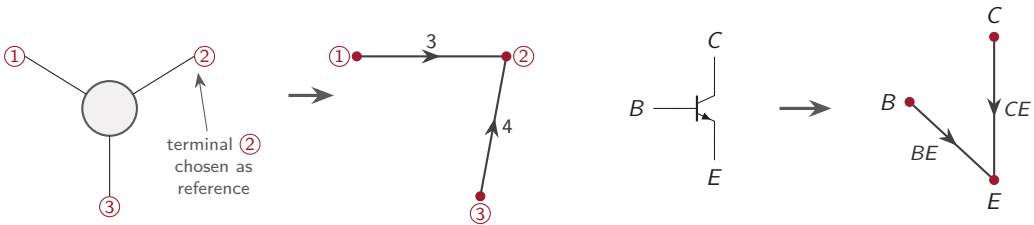


Why $k-1$? KCL on a surface around the element: the k terminal currents sum to zero, so only $k-1$ are free. By KVL, the $k-1$ voltages to the reference fix all the others.

Elements with more than two terminals

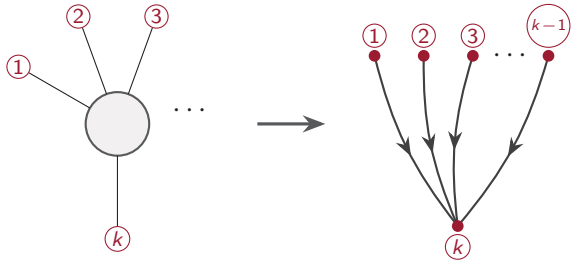
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Why $k-1$? KCL on a surface around the element: the k terminal currents sum to zero, so only $k-1$ are free. By KVL, the $k-1$ voltages to the reference fix all the others. A transistor: emitter as reference, two branches.

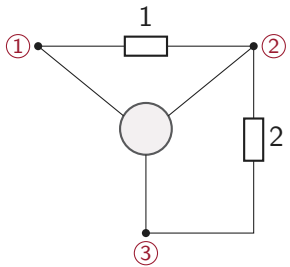
The general case



a k-terminal element $k-1$ **branches, all to terminal** k

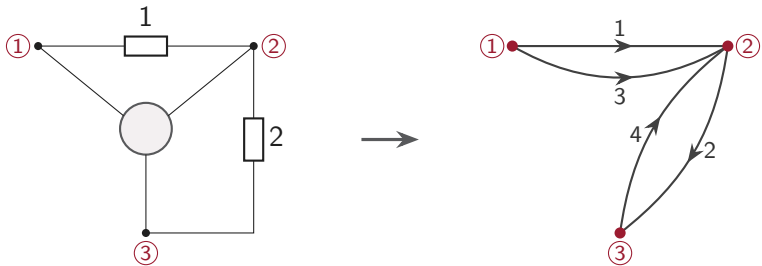
A dependent source needs nothing new: it is already two branches — the branch that senses the controlling variable, and the source branch.

Example: a circuit with a 3-terminal element



Elements 1 and 2 have two terminals; the circle has three. Find the graph.

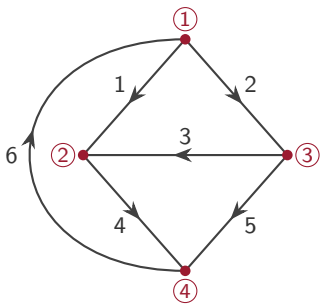
Example: a circuit with a 3-terminal element



Elements 1, 2: one branch each. 3-terminal element, reference terminal at ②: branches 3 (①→②) and 4 (③→②).

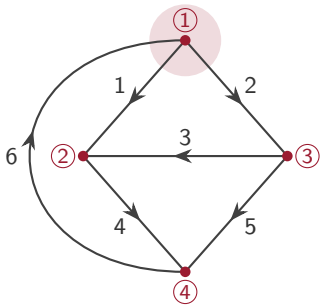
$n = 3$, $b = 4$. Two branches may join the same pair of nodes.

KCL on a circuit graph



Current along its arrow. At each node:
leaving $+$, entering $-$, sum $= 0$.

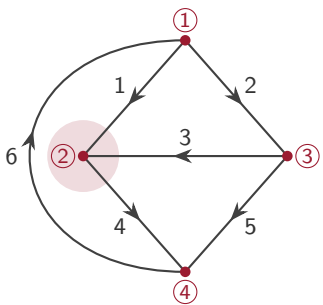
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Current along its arrow. At each node:
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$$①: \quad i_1 + i_2 - i_6 = 0$$

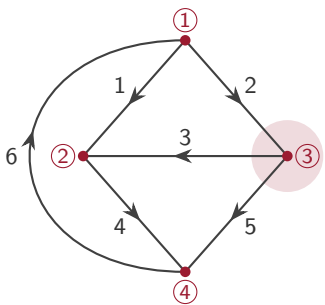
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$$\begin{aligned} ①: &\quad i_1 + i_2 - i_6 = 0 \\ ②: &\quad -i_1 - i_3 + i_4 = 0 \end{aligned}$$

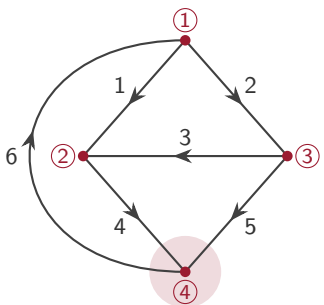
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\end{aligned}
$$

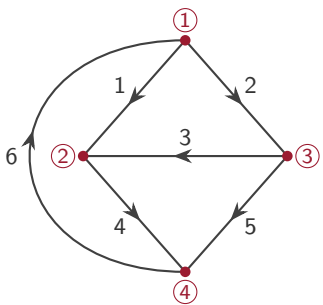
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$$

Only the graph was used. Same graph $\Rightarrow$ same KCL equations, whatever the elements are.

From equations to a matrix

Only $+1$, -1 and 0 appear. Rows: nodes. Columns: branches.

$$\begin{array}{c} \\ ① \\ ② \\ ③ \\ ④ \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & -1 \\ -1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & -1 & 1 \end{bmatrix} \end{array} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \\ i_6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \Longleftrightarrow \quad A_a \mathbf{i} = \mathbf{0}$$

A matrix that stores the graph: a circuit matrix. This one is the first and most basic.

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$$
\begin{array}{c}
 \\
① \\
② \\
③ \\
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= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}
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$$

A row is a node: row ① times $\mathbf{i}$ is KCL at ①.

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A column is a branch: one $+1$ (leaves) and one -1 (enters). Column 3: from ③ to ②.

The incidence matrix

Incidence matrix A_a $(n \times b)$

$$a_{ij} = \begin{cases} +1 & \text{if branch } j \text{ leaves node } i, \\ -1 & \text{if branch } j \text{ enters node } i, \\ 0 & \text{otherwise.} \end{cases}$$

With $\mathbf{i} = [\, i_1 \ \cdots \ i_b \,]^T$, the branch current vector: KCL at every node is $A_a\mathbf{i} = \mathbf{0}$.

- The subscript a: *all* nodes have a row.

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- Reversing an arrow flips the sign of that column — nothing else changes.

Review: linear dependence

Linear dependence

$\mathbf{v}_1, \ldots, \mathbf{v}_k$ (same size) are linearly dependent if there are real $\alpha_1, \ldots, \alpha_k$, **not all zero**, with $\alpha_1 \mathbf{v}_1 + \cdots + \alpha_k \mathbf{v}_k = \mathbf{0}$. Otherwise they are linearly independent.

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$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

independent: $[\alpha_1 \quad -\alpha_2]^T = \mathbf{0}$ only if both are zero.

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$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}$$

dependent: $\mathbf{v}_1 - 2\mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$, so $\mathbf{v}_3 = 2\mathbf{v}_2 - \mathbf{v}_1$.

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Rows of M dependent $\Rightarrow$ one equation of $M\mathbf{x} = \mathbf{0}$ follows from the others.
Number of independent rows: the rank. All rows independent: full row rank.

The rows of A_a are always dependent

Every column has one $+1$ and one -1, so

$$\text{row } \textcircled{1} + \text{row } \textcircled{2} + \cdots + \text{row } \textcircled{n} = [0\ 0\ \cdots\ 0]$$

— dependence with every $\alpha = 1$. One KCL equation is always redundant.

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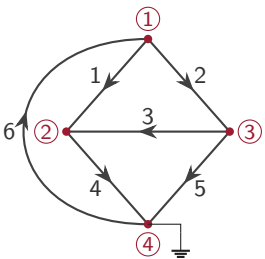
Reduced incidence matrix A $((n-1) \times b)$

Choose a reference node (ground), arbitrarily. Delete its row from A_a.

The reference node is also the common reference for node voltages.

Example: the reduced incidence matrix of $\mathcal{N}$

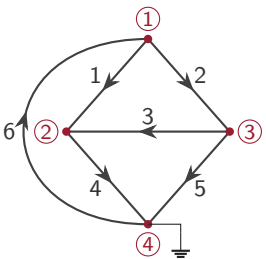
Reference node: ④.



$$A_a = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & -1 \\ -1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & -1 & 1 \end{bmatrix}$$

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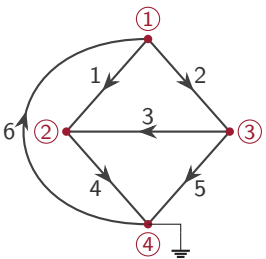
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$$A = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & -1 \\ -1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \end{bmatrix} \qquad (3 \times 6)$$

Example: the reduced incidence matrix of $\mathcal{N}$



Reference node: ④.

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Rows independent: columns 6, 4, 5 each have a single nonzero entry, in rows ①, ②, ③.

Deleting the row loses nothing

No equation is lost

$A\mathbf{i} = \mathbf{0} \Rightarrow A_a\mathbf{i} = \mathbf{0}$: the deleted row is minus the sum of the rows that were kept.

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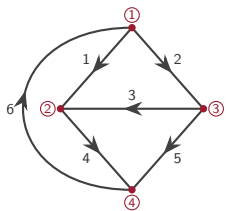
No part of the graph is lost

Each column of A_a sums to zero, so the deleted row can be rebuilt from A.
A column of A with a single nonzero entry is a branch that touches the reference node.

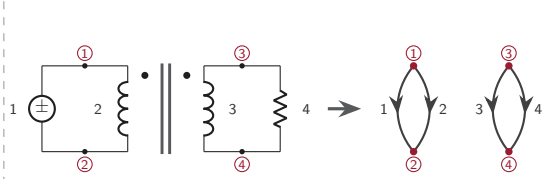
Connected graphs

Connected graph

Any two nodes are joined by a path of branches, ignoring the arrow directions.
Otherwise the graph falls into separate pieces.



graph of $\mathcal{N}$: connected



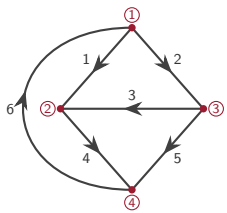
a transformer circuit: two pieces

The windings are coupled only through the core: no branch joins the two sides.

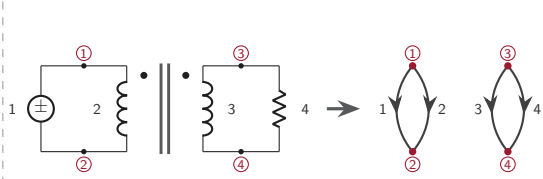
Connected graphs

Connected graph

Any two nodes are joined by a path of branches, ignoring the arrow directions.
Otherwise the graph falls into separate pieces.



graph of $\mathcal{N}$: connected

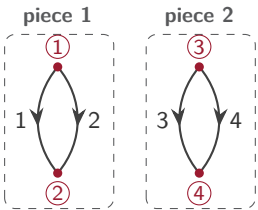


a transformer circuit: two pieces

The windings are coupled only through the core: no branch joins the two sides.
Also: two parts that interact only through a dependent source that senses in one part and acts in the other — an isolation amplifier, an optocoupler.

Graphs in separate pieces

Handle each piece on its own, then stack the reduced matrices.

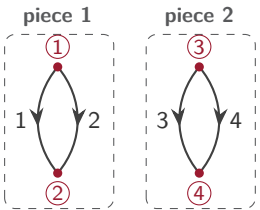


the transformer circuit

	incidence matrix	reference	reduced
piece 1			
piece 2			

Graphs in separate pieces

Handle each piece on its own, then stack the reduced matrices.



the transformer circuit

incidence matrix **reference** **reduced**

piece 1

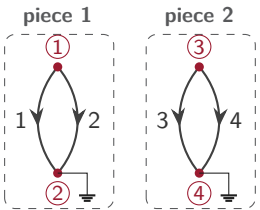
$$\begin{array}{c} \\ ① \\ ② \end{array}\begin{array}{c} \begin{array}{cc} 1 & 2 \end{array} \\ \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \end{array}$$

piece 2

$$\begin{array}{c} \\ ③ \\ ④ \end{array}\begin{array}{c} \begin{array}{cc} 3 & 4 \end{array} \\ \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \end{array}$$

Graphs in separate pieces

Handle each piece on its own, then stack the reduced matrices.

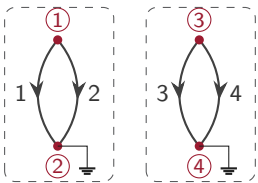


the transformer circuit

	incidence matrix	reference	reduced
piece 1	$\begin{array}{c} \\ ① \\ ② \end{array}\begin{bmatrix} 1 & 2 \\ 1 & 1 \\ \cancel{1} & \cancel{1} \end{bmatrix}$	delete row ②	$A_1 = ① \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$
piece 2	$\begin{array}{c} \\ ③ \\ ④ \end{array}\begin{bmatrix} 3 & 4 \\ 1 & 1 \\ \cancel{1} & \cancel{1} \end{bmatrix}$	delete row ④	$A_2 = ③ \begin{bmatrix} 3 & 4 \\ 1 & 1 \end{bmatrix}$

Graphs in separate pieces

Handle each piece on its own, then stack the reduced matrices.



the transformer circuit

incidence matrix **reference** **reduced**

piece 1

$$\begin{matrix} & 1 & 2 \\ ① & 1 & 1 \\ ② & 1 & 1 \end{matrix}$$

delete row ② $A_1 = ①$ $\begin{matrix} & 1 & 2 \\ & 1 & 1 \end{matrix}$

piece 2

$$\begin{matrix} & 3 & 4 \\ ③ & 1 & 1 \\ ④ & 1 & 1 \end{matrix}$$

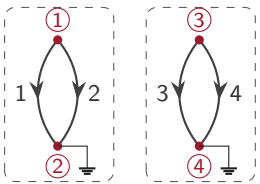
delete row ④ $A_2 = ③$ $\begin{matrix} & 3 & 4 \\ & 1 & 1 \end{matrix}$

Stack: $A = \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix} = \begin{matrix} ① \\ ③ \end{matrix} \begin{matrix} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \\ \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \end{matrix}$

$i_1 + i_2 = 0, \quad i_3 + i_4 = 0$

Graphs in separate pieces

Handle each piece on its own, then stack the reduced matrices.



the transformer circuit

incidence matrix **reference** **reduced**

piece 1 $\begin{matrix} \\ ① \\ ② \end{matrix}\begin{bmatrix} 1 & 2 \\ 1 & 1 \\ \cancel{1} & \cancel{1} \end{bmatrix}$ delete row ② $A_1 = \begin{matrix} ① \end{matrix}\begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$

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p pieces: p reference nodes, so A has $n - p$ independent rows. Here $4 - 2 = 2$.

KCL of a circuit graph

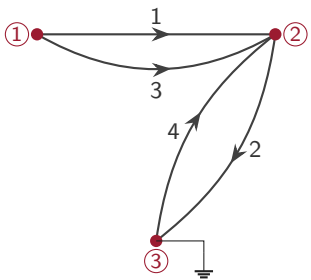
$$A\mathbf{i} = \mathbf{0}$$

$A \in \mathbb{R}^{(n-1)\times b}$	reduced incidence matrix
$\mathbf{i} \in \mathbb{R}^{b}$	branch current vector

For a connected graph: $n-1$ independent KCL equations,
and every other KCL equation of the circuit follows from them.

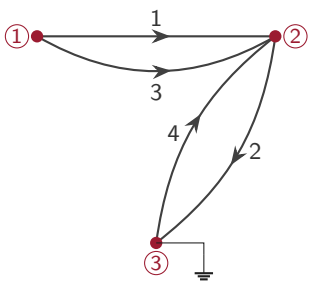
Separate pieces: one reference node per piece, $n-p$ rows.

Example: KCL for the circuit with a 3-terminal element



Reference node: ③.

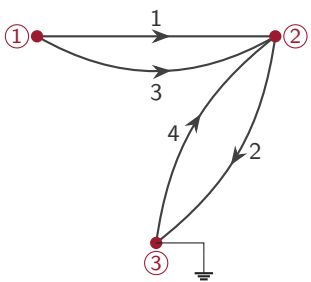
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$$A_a = \begin{array}{c} \\ ① \\ ② \\ ③ \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 1 & 0 & 1 & 0 \\ -1 & 1 & -1 & -1 \\ 0 & -1 & 0 & 1 \end{bmatrix} \end{array}$$

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Example: KCL for the circuit with a 3-terminal element

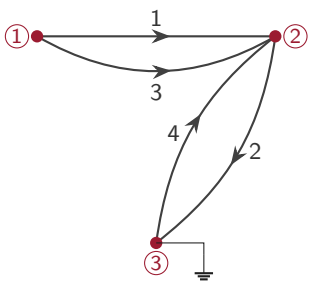


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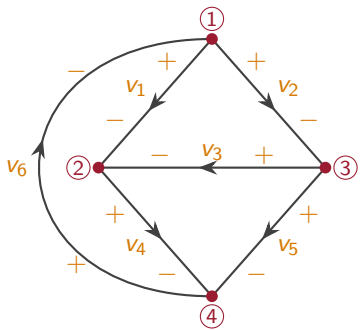
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$$A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ -1 & 1 & -1 & -1 \end{bmatrix}$$

$$i_1 + i_3 = 0, \qquad -i_1 + i_2 - i_3 - i_4 = 0$$

Columns 1 and 3 are equal: same nodes, same direction. Reference terminal (②) and reference node (③): two separate choices.

Branch voltages and node voltages

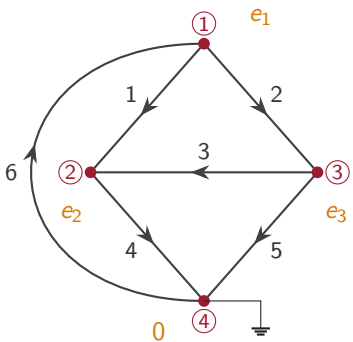


Branch voltage v_k



$+$ at the tail: i_k enters at $+$. $b = 6$

Branch voltages and node voltages



Branch voltage v_k



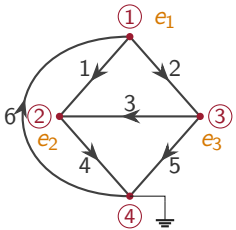
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Node voltage e_k

Node k with respect to the reference.
Reference: $e = 0$. $n - 1 = 3$

v_{pq}: node p $(+)$ w.r.t. node q $(-)$.
$e_1 = v_{14}$, $e_2 = v_{24}$, $e_3 = v_{34}$

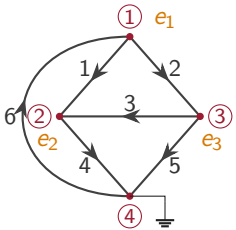
KVL on a circuit graph



Branch 1, from ① to ② — go through the reference ④:

$$v_1 = v_{12} = v_{14} + v_{42} = v_{14} - v_{24} = e_1 - e_2$$

KVL on a circuit graph

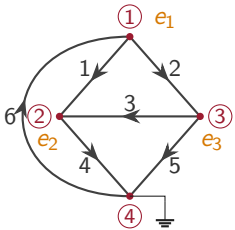


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KVL on a circuit graph



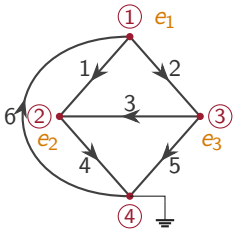
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$$\left.\begin{array}{ll} v_1 = e_1 - e_2 & v_4 = e_2 \\ v_2 = e_1 - e_3 & v_5 = e_3 \\ v_3 = e_3 - e_2 & v_6 = -e_1 \end{array}\right\} \Longrightarrow \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{bmatrix}}_{A^T} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}$$

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This matrix is A^T: its row 1, $[\,1 \;\; -1 \;\; 0\,]$, is column 1 of A — and so on.

KVL in matrix form

$$\mathbf{v} = A^T \mathbf{e}$$

$\mathbf{v} \in \mathbb{R}^b$ branch voltage vector

$\mathbf{e} \in \mathbb{R}^{n-1}$ node voltage vector

The same A gives KCL ($A\mathbf{i} = \mathbf{0}$) and KVL.

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- KVL puts no constraint on $\mathbf{e}$: any $\mathbf{e}$ satisfies KVL around every loop. Loop 1, 3, 2: $v_1 - v_3 - v_2 = (e_1 - e_2) - (e_3 - e_2) - (e_1 - e_3) = 0$

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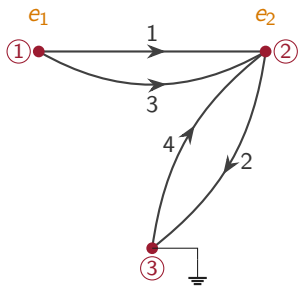
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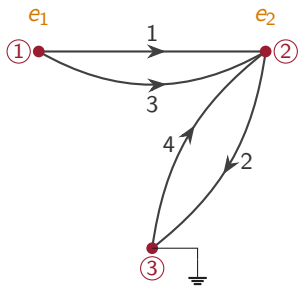
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- So node voltages are good unknowns: only $n-1$, free, KVL built in — the idea behind node analysis.

Example: KVL for the circuit with a 3-terminal element



$A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ -1 & 1 & -1 & -1 \end{bmatrix}$ (reference ③), so two node voltages.

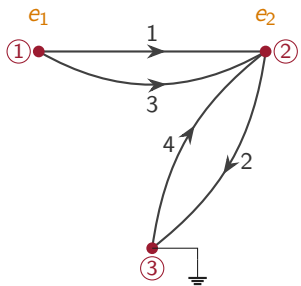
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Branches 1 and 3 join the same nodes: same voltage.
Branch 4 goes from the reference to ②: $v_4 = 0 - e_2$.

Tellegen's theorem

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In any circuit, with associated references,

$$\mathbf{i}^T \mathbf{v} = \sum_{k=1}^{b} v_k \, i_k = 0 \qquad \text{at every instant } t.$$

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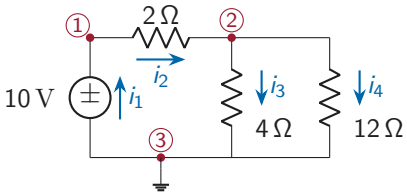
In any circuit, with associated references,

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$v_k i_k$ = power absorbed by branch k. Total absorbed power = 0:
conservation of power — what some branches deliver, the others absorb.

Example: checking Tellegen's theorem

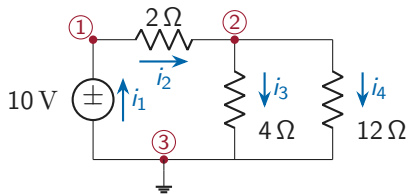
Find v_k and i_k of every branch with node voltages, then check $\sum_k v_k i_k = 0$.



reference ③; + of v_k at the tail of each arrow

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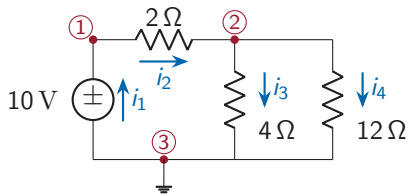


Node voltages: $e_1 = 10\,\mathrm{V}$ (source), e_2 unknown.

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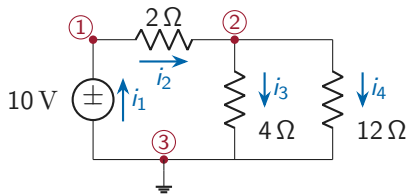
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KCL at ②: $-\dfrac{e_1 - e_2}{2} + \dfrac{e_2}{4} + \dfrac{e_2}{12} = 0 \;\Rightarrow\; e_2 = 6\,\mathrm{V}$

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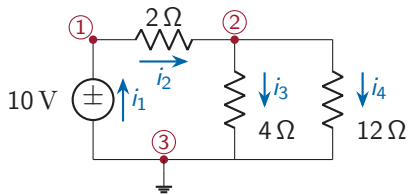
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$\mathbf{v} = A^T\mathbf{e}$**:** $v_1 = -e_1 = -10$, $v_2 = 4$, $v_3 = v_4 = 6\,\mathrm{V}$

Ohm, KCL at ①: $i_1 = i_2 = 2$, $i_3 = 1.5$, $i_4 = 0.5\,\mathrm{A}$

Example: checking Tellegen's theorem

Find v_k and i_k of every branch with node voltages, then check $\sum_k v_k i_k = 0$.



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Ohm, KCL at ①: $i_1 = i_2 = 2$, $i_3 = 1.5$, $i_4 = 0.5\,\mathrm{A}$

reference ③; + of v_k at the tail of each arrow

branch	element	v_k (V)	i_k (A)	$v_k i_k$ (W)
1	source	−10	2	−20
2	2 Ω	4	2	8
3	4 Ω	6	1.5	9
4	12 Ω	6	0.5	3
			sum	0

$v_1 < 0$: its + (tail) is at ③.
The source delivers 20 W; the resistors absorb $8 + 9 + 3 = 20$ W.

Summary

- $2b$ equations: b **element** equations and b **interconnection** equations (KCL, KVL). The second group depends only on the graph.
- The graph: number the branches, give each an arbitrary arrow, number the nodes. A k-terminal element becomes $k-1$ branches.
- **Incidence matrix** A_a ($n \times b$): $+1$ leaves, -1 enters. KCL: $A_a\mathbf{i} = \mathbf{0}$. Its rows always sum to zero.
- **Reduced incidence matrix** A: delete the reference node's row. For a connected graph, $A\mathbf{i} = \mathbf{0}$ gives exactly the $n-1$ independent KCL equations.
- **KVL**: $\mathbf{v} = A^T\mathbf{e}$ — tail minus head. No constraint on the node voltages $\mathbf{e}$.
- **Tellegen**: in any circuit $\mathbf{i}^T\mathbf{v} = \sum v_k i_k = 0$ — conservation of power.

Next lecture

Planar graphs and meshes

The mesh matrix M: KVL as $M\mathbf{v} = \mathbf{0}$, and mesh currents with $\mathbf{i} = M^T \mathbf{i}_M$ — the counterpart of $\mathbf{v} = A^T \mathbf{e}$.

Questions?